

# Yan Wang

Hongyi Honors Class, College of Life Sciences, Wuhan University, Wuhan, 430072, China

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## EDUCATION

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**Wuhan University**, Wuhan, China *China #8 · U.S. News #85*

*Sep. 2023-Present*

B.S. in Biological Science; **GPA: 3.91/4.00; Rank: 1/33; Major GPA: 3.97/4.00**

Selected coursework: Cell Biology(95); Molecular Biology(96); Genetics(96); Biochemistry(95)

**Tsinghua University High School**, Beijing, China

*Sep. 2020-Jun. 2023*

National College Entrance Examination, **Biology:100/100**

## SELECTED RESEARCH

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**Adams Center for Parkinson's Disease Research, Yale School of Medicine**

*Jul. 2026-Present*

Visiting Undergraduate Researcher, mentored by Prof. Clemens Scherzer and Dunhui Li

**Project title: Functional Dissection of the PM20D1 Locus in Parkinson's Disease**

- Investigate how PM20D1 perturbation affects mitochondrial function and cellular redox homeostasis in human iPSC models using antisense oligonucleotides (ASOs) and gene-editing approaches.
- Characterize the subcellular localization of the lncRNA PM20D1-AS1 and investigate its potential regulation of PM20D1 expression, while exploring candidate molecular interaction partners.
- Independently develop RNA-seq analysis workflows to profile 22 annotated PM20D1-AS1 transcripts, identifying preliminary evidence of increased relative abundance of specific isoforms in astrocytes and oligodendrocytes from patients with Parkinson's disease.

**Medical Research Institute, Wuhan University**

*Jan. 2025-July. 2026*

Joint lab of Prof. Yanxun Yu and Prof. Younghan Jin

**Project title: INS-1-Dependent Neuropeptidergic Tuning of Sensory Valence**

- Lead a *C. elegans* neuroscience project investigating how insulin-like neuropeptide(INS-1) signaling assigns negative valence to ethanol under high-salt conditions.
- Discover that *ins-1* loss completely reverses high-salt ethanol avoidance into attraction.
- Map distributed INS-1-releasing nodes by combining cell-specific *ins-1* deletion, rescue, and neuronal ablation, implicating ASER, AIY, AIZ, and AIB in ethanol avoidance.
- Demonstrate ASER-AIY circuit dysfunction using calcium imaging, showing a right-shifted ASER ethanol response range and weakened AIY negative calcium transients in *ins-1* mutants.
- Reveal ASER-derived INS-1 as a major regulator of ASER sensory tuning. Integrate ethanol dose-dependent behavioral assays with physical modeling of diffusion, supporting a quantitative model in which INS-1 tunes ethanol sensitivity rather than causing simple sensory loss.
- Proposed NTR-1 as a candidate noncanonical INS-1 receptor by integrating pathway mutant analysis, *ntr-1* phenotyping, molecular docking and MD Simulation.
- Developed a circuit-level model in which distributed INS-1 release, potentially mediated by NTR-1, calibrates ASER sensory gain and downstream AIY output to code ethanol valence under high-salts.

**Project title: Performance Optimization of the POST-IT Proximity Tagging System**

- Optimize POST-IT, a Pupylation-based non-diffusive proximity-labeling platform for live-cell-compatible small-molecule target identification.
- Engineer PafA ligase to reduce the major kinetic and substrate constraints of the original POST-IT system, including long labeling time and high SBP-Pup (E) requirement.
- Build a yeast-surface-displayed Pep1-MATH weak-interaction selection system to couple PafA catalytic activity with FACS-based enrichment of improved variants.
- Generate a  $\sim 10^6$ -scale PafA mutant library by error-prone PCR and performed iterative directed evolution under increasingly stringent labeling conditions.

- Combine FACS and NGS-guided mutation analysis to obtain high-performing PafA variants.
- Improve labeling from 18 h / 40  $\mu$ M to 2 h / 10  $\mu$ M, increasing threshold-positive cells from 0.5% to 27.2% and achieving >40-fold apparent performance enhancement.

College of Life Science, Wuhan University

Nov. 2024-Oct. 2025

**Project title: LOGIC -Lasting Optimized Genetic Input Calculator, 2025 IGEN Gold Medal**

- Original proposer and core system designer of LOGIC, which utilizes quorum-sensing-mediated intercellular communication as programmable spatial “wiring” for colony-level bio-computation.
- Design a diffusion-based biocomputing architecture in which orthogonal AHL signaling channels, engineered sender-receiver colonies, and spatial concentration gradients are integrated to implement modular genetic logic beyond single-cell circuit constraints.
- Build and characterized high-pass and band-pass response modules to encode Boolean operations, construct AND, OR, XOR, multi-input gates, and cascaded logic operations on solid media.
- Develop a serial biological adder framework by coordinating input, output, and carry signals, advancing the system from isolated logic gates toward executable biological arithmetic.
- Integrate optogenetic refresh modules, including split-VVD AiiA for light-controlled AHL degradation and LOVdeg-tagged output proteins for signal reset.

## PUBLICATIONS, CONFERENCES & FUNDING

- **Wang, Y.**, Miller, L., Estrada, M., & Li, D. Mitochondrial dysfunction and  $\alpha$ -synuclein pathology in Parkinson’s disease: A bidirectional pathogenic loop and therapeutic opportunities. *Frontiers in Molecular Neuroscience*. **Review manuscript submitted.**
- **Wang, Y.**, Zhang, J., Huang, X., Ruan, Y., & Yu, Y. Distributed INS-1 signaling through the candidate noncanonical receptor NTR-1 tunes the ASER–AIY axis to drive high-salt ethanol avoidance in *Caenorhabditis elegans*. *Neuroscience 2026, Society for Neuroscience*. **Conference abstract accepted.**
- Funding proposal: Multimodal monitoring of reversible hypometabolic states. Unitree Robotics, Young Talent Sponsorship Program. Application under review (September 2026).

## TECHNICAL SKILLS

- Microbial and model organism handling: *E. coli*, yeast, iPSC, and *C. elegans*.
- Molecular cloning, RNA experiments (e.g., qPCR, RNAscope, RNA pull-down.), protein expression and purification, Western blot, co-immunoprecipitation, FACS, and biochemical assays.
- Microinjection, crosses, calcium imaging, optogenetic manipulation, and behavioral assays in *C. elegans*.
- Bioinformatics analysis (e.g., snRNA-seq) and structural analysis (e.g., MD simulations).

## ACADEMIC ACTIVITIES

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|---------------------------------------------------------------------------------------|--------------|
| WHU Delegate to the National “Top-Tier Talent 2.0” Biology Forum, Tsinghua University | May 2025     |
| “Frontiers of Life Sciences” International Summer School, Westlake University         | Jul–Aug 2025 |
| Neuroscience & Cognitive Science Summer Training Program, Peking University           | Jul 2026     |

## HONORS & AWARDS

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|-----------------------------------------------------------------------------------------|-----------|
| Gold Medal, International Genetically Engineered Machine Competition (iGEM);            | Nov. 2025 |
| Gold Prize , Regular Track, SynBio Challenges (an international contest)                | 2025&2026 |
| Gold Prize , Synthetic Cell Track, SynBio Challenges                                    | Aug. 2026 |
| Gold Prize , Protein Design Track, SynBio Challenges                                    | Aug. 2026 |
| Gold Prize , Bacterial GameTrack, SynBio Challenges                                     | Aug. 2026 |
| Liu Daoyu Creative Learning Scholarship (¥8000, awarded to 24 students university-wide) | Jan. 2026 |
| Hongyi Honors College’s Annual Top Ten Science Advances Award                           | Oct. 2025 |
| Westlake University Rising Star Award (¥30000, awarded to 9 from 300 candidates.)       | Aug. 2026 |

First-Class Outstanding Freshman Scholarship (2023); Outstanding Student Scholarship (5%, 2024 & 2025); Outstanding Student (5%, 2024 & 2025); Outstanding Student Leader (2024 & 2025)